LESSON PLAN

GUIDED WEAPON (37)

Class	Air Agniveer Apprentice (Engineering)	**Lesson Plan No.**	03
Total No. of Lesson Plans	6	**Time Allotted for the Session**	80 min

PART A: GUIDELINES / GENERAL INSTRUCTIONS FOR INSTRUCTORS

1. Topic	External ballistics –motion in vacuum, motion in air, formula for air resistance, ballistic coefficient and limiting velocity
2. Total No. of Sessions for this Topic	01 / 01
3. Sequence of this Session	01 / 01
4. Syllabus Extract	External ballistics –motion in vacuum, motion in air, formula for air resistance, ballistic coefficient and limiting velocity
5. Specific Objectives	37.1.1.1 Trainee will be able to explain about various branches of ballistics

6. List of Training Aids

Ser	Description	Remarks
(a)	BR/Text Book/Docket	Refer to the master syllabus reference list
(b)	Wall Charts	Nil
(c)	Films/Duration	Nil
(d)	PPT	Objective: teach the syllabus extract for this lesson
(e)	Models	Nil
(f)	CBT	Nil

7. Practical in Lab/Equipment Rooms/Simulators	Review of trajectory charts; Safety: No live ammunition; Supervision: Instructor monitors chart interpretation
8. Existing Knowledge of the Class	As per course prerequisites

PART B: DEVELOPMENT OF LESSON

9. Checking of Previous Knowledge / Introduction	Time: 10 min

(a) What are the four main branches of ballistics?
(b) Define external ballistics in one sentence.
(c) List two factors that affect projectile motion in air.

10. Development of Main Lesson

Sub	Content	Method	Trg Aids	Time
(a)	• Definition of external ballistics • Motion in vacuum vs motion in air • Factors affecting air resistance • Formula for air resistance • Ballistic coefficient definition	L & Ds	PPT, IB	15 min
(b)	• Concept of limiting velocity • Derivation of limiting velocity formula • Relationship between drag and velocity • Effect of shape on ballistic coefficient • Comparison of vacuum and air trajectories	L & Ds	PPT, PDF	20 min
(c)	• Application of ballistic coefficient • Calculation of limiting velocity • Trajectory deviation in air • Importance of air density • Summary of external ballistics concepts	Practical	IB, PDF	15 min

11. Summary and Evaluation

Method	Trg Aids	Time
Q & A	IB	15 min

(a) Explain how air resistance differs from vacuum motion.
(b) Describe the significance of the ballistic coefficient in weapon performance.

12. Home Assignment and Follow Up	Prepare a brief note on how the concepts of external ballistics learned in this lesson will assist in understanding terminal ballistics and damage effects covered in LP 04.
Time	5 min

PART C: PREPARATION FOR TOMORROW (PFT) DIRECTIVES FOR TRAINEES

Ser	Syllabus for PFT	References	Related LPs
1	Terminal ballistics, type of damage, perforation effect, critical velocity. Experimental ballistics – measurement of pressure and velocity	Refer to the master syllabus reference list	LP 04

PART D: PERSONAL EXPERIENCES (PE) OF INSTRUCTOR

Ser	Item	Remarks
(a)	Adequacy of time	
(b)	Suitability of training material/aids and additional aids required	
(c)	Doubts/intelligent questions raised by trainees	
(d)	Difficulties faced by instructor	
(e)	Other experience	

Instructor	**HoF (E/L)**
Signature: _____	Signature: _____

PART E: REVIEW SHEET

Date	Instructor's Comment	HOF (E/L)	Oi/C TDEC Remarks

Prepared by / Approved by

Prepared by	Approved by
Name, Rank, P No.	Name, Rank, P No.
Nov 26	Nov 26

Queries for Instructor

1. Confirm the master syllabus reference book/BR for this subject.
2. Confirm the class existing knowledge and prerequisites.
3. Confirm the session duration and sequence; 80 min and 01/01 were assumed.
4. Prepared-by/approved-by names, ranks, P numbers and dates must be filled before approval.
5. Confirm any training aids available beyond L & Ds, IB, PDF and PPT.

Timing Check

Introduction 10 + Main 50 + Summary 15 + Home 5 = 80 min